Short Vitae

Personal Information

Name	Seungchan Woo
Company	Protocol Engineering Lab (Sejong University)
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EDUCATION

Sejong University, Graduate School, Seoul, Republic of Korea
Ph.D. Student in Computer and Information Security, September 2025 - Present
Ph.D. Student in Convergence Engineering for Intelligent Drone, September 2025 - Present
Supervisor: Prof. Jong-Hyouk Lee

GPA: N/A (in progress)

Sejong University, Graduate School, Seoul, Republic of Korea
Master of Science in Computer and Information Security, August 2025
Master of Science in Convergence Engineering for Metaverse, August 2025
Supervisor: Prof. Jong-Hyouk Lee

GPA: 4.50 / 4.50, equivalent to 100 / 100

Sejong University, Seoul, Republic of Korea
Bachelor of Science in Computer and Information Security, August 2023

GPA: 3.54 / 4.50, equivalent to 89.4 / 100

EXPERIENCE

TTA/TC 4/PG 428 (ICT Convergence/Drone System) February 2025 - Present

- Contributed to domestic standardization activities for anti-drone systems as a committee member.
- Contributed to the development of domestic standards for anti-drone system interfaces and data specifications.

ISO/TC 20/SC 16/WG 8 (Aircraft and space vehicles/Uncrewed aircraft system/Counter UAS) June 2026 - Present

- Contributed to international standardization activities for Counter UAS systems as a working group member.
- Contributed to the development of ISO 25461 on functional requirements for Counter UAS detection, tracking, and identification.
- Contributed to the development of a proposed ISO standard on data exchange interfaces for Counter UAS detection, tracking, and identification.

ISO/TC 20/SC 16/WG 10 (Aircraft and space vehicles/Uncrewed aircraft system /Cybersecurity) July 2026 - Present

- Participating in international standardization activities on cybersecurity for uncrewed aircraft systems as a working group member.
- Participated online in the WG 10 meeting during the 21st ISO/TC 20/SC 16 plenary in Brussels, Belgium, in July 2026.

Visiting Researcher, Ruhr University Bochum, Bochum, Germany

December 2025 - May 2026 (six months) | Host: Prof. Tarik Taleb

- Conducted research on distributed ledger-based trust models for UxNB services in 6G Open RAN environments.
- Analyzed Open RAN security threats and investigated trust architectures and attack scenarios for UxNB deployment.
- Collaborated with the host research team on cybersecurity, privacy protection, and trust management for next-generation networks.

Visiting Researcher, University of Oulu, Oulu, Finland September 2023

- Conducted attack experiments on the University of Oulu's Open Source 5GTN testbed as part of Korea-Finland joint research on security datasets.
- Collaborated through weekly meetings with Prof. Mika Ylianttila's research group and the 5GTN team on attack experiments and data collection.
- Presented the attacks performed on Open Source 5GTN and data collection methods at the joint research meeting on September 22, 2023; participated in a 6G research workshop with VTT and ETRI.

SKILLS AND ACTIVITIES

Researcher, Protocol Engineering Lab, Sejong University, Seoul, Republic of Korea July 2021 - Present

- Conducted research on cybersecurity, privacy protection, blockchain/DLT, 6G network security, Open RAN security, drone security, and anti-drone system standardization.
- Participated in national R&D projects funded by KEIT, IITP, KISA, NSR, ETRI, NRF, KOCCA and KISTI.
- Contributed to domestic and international standardization activities related to anti-drone systems, drone cybersecurity and 6G security.
- Published and presented research papers on anti-drone systems, UxNB security, software copyright protection, and 5G-Advanced/6G security.

Project: Study of International Cybersecurity Standards for the Secure Operation of AI-based Drones

April 2026 - Present

- Conducting research on cybersecurity requirements and standardization strategies for the secure operation of AI-based drones.
- Supporting the development of international standards for AI-based drone security, safety, and secure data exchange.
- Analyzing cybersecurity threats, operational risks, and security requirements for drone systems in intelligent and autonomous environments.

Project: Development of International Standards for Interoperability of Counter-UAS July 2025 - Present

- Conducting research on interoperability requirements for counter-UAS and anti-drone integrated systems.
- Developing international standardization items related to counter-UAS architectures, data exchange interfaces, and information elements.
- Contributing to the development of domestic and international standards for detection, tracking, identification, and mitigation interfaces in anti-drone systems.
- Analyzed anti-drone system frameworks and interface specifications to support standard based interoperability.

Project: Training Global Talent for Copyright Protection and Management of On-Device AI Models

April 2025 - Present

- Conducting research on copyright protection and management technologies for on-device AI models.
- Analyzing technical and policy requirements for protecting AI model ownership, licensing, and distribution.
- Supporting international cooperation and research activities related to AI model copyright protection.

Project: Research and International Cooperation on Trust Model-based Intelligent Breach Response Technology in 6G Open Network Environment July 2024 - Present

- Conducting research on trust models, intelligent breach response, and security mechanisms for 6G open network environments.
- Analyzing security threats and attack models in Open RAN and 6G network architectures.
- Supporting international cooperation and standardization research for secure and trustworthy 6G networks.

Project: Study of Anti-drone System Standards Conformance Testing April 2025 - October 2025

- Conducted research on conformance testing frameworks for anti-drone system standards.
- Analyzed test methodologies, testing procedures, and conformity assessment models for anti-drone systems.
- Supported the development of standard-based verification approaches for anti-drone system interoperability and conformance testing.
- **Seungchan Woo** and Jong-Hyouk Lee, "Anti-Drone System Framework," TTA Journal, vol. 222, pp. 34–36, December 2025.
- **Seungchan Woo**, Jinsue Lee, Haneul Lee and Jong-Hyouk Lee, "A Study on Interface and Data Specification Standardization for Anti-Drone System Interoperability," 2025 Society for Standards Certification and Safety Fall Conference, October 2025.

Project: Study of Requirements Analysis and Standardization of Data Specifications for Interoperability of Anti-Drone System April 2024 - October 2024

- Researched data specification requirements for interoperability among anti-drone system components.
- Analyzed anti-drone system architectures, interface requirements, and information exchange models.
- Contributed to the development of domestic standards for anti-drone system.

Project: Collection of 6G Infrastructure Attack Traffic and Development of Datasets through International Collaboration August 2023 - November 2023

- Supported research on attack traffic collection for 6G infrastructure environments.
- Analyzed network attack scenarios and dataset development methods.
- Contributed to international collaborative research on 6G security datasets.

Project: Study of Security Standardization Trends for Drone and Anti-drone Systems and Establishment of Strategies for Propulsion April 2023 - October 2023

- Analyzed domestic and international security standardization trends for drone and anti-drone systems.
- Researched cybersecurity requirements and standardization strategies for unmanned systems.
- Supported the establishment of standardization strategies for drone and anti-drone security.

Project: Study of International Standardization for Edge-based Virtualization Security Technology in 6G Environments January 2023 - December 2023

- Analyzed security threats and requirements for edge-based virtualization technologies in 6G environments.
- Researched Open RAN security issues and attack models.
- Supported the establishment of international standardization strategies for 6G and Open RAN security

Project: Development of Software Copyright Application Technology for Fair Trade and Distribution April 2022 - November 2023

- Conducted research on blockchain-based software copyright protection, licensing, and distribution technologies.
- Developed research concepts for decentralized identifiers and license management in software copyright transactions.
- Analyzed blockchain and DLT technologies for supporting fair software copyright transactions.
- **Seungchan Woo**, Taeyang Lee and Jong-Hyouk Lee, "Decentralized Identity System for Software Copyright Transactions," KIISC Winter Conference, December 2023.
- **Seungchan Woo**, Taeyang Lee and Jong-Hyouk Lee, "Decentralized Identifier System for Software Copyright Transfer and License Management," IEEE International Conference on Metaverse Computing, Networking and Applications (IEEE MetaCom 2023), pp. 692–693, June 2023.

Project: Research on Foundational Technologies for 6G Autonomous Security by Design to Guarantee Constant Quality of Security July 2021 - December 2024

- Conducted research on 6G autonomous security, security-by-design, and network security management.
- Analyzed security issues in network virtualization, container networking, and 6G infrastructure environments.

- Supported research on attack traffic collection, network modeling, and security dataset development for 6G infrastructure.
- **Seungchan Woo** and Jong-Hyouk Lee, "UxNB Security Testbed Configuration for 5G-Advanced," ASK 2024, May 2024.
- **Seungchan Woo**, Jaehyoung Park, Soonhong Kwon, Kyungmin Park, Jonghyun Kim and Jong-Hyouk Lee, "Simulation of Data Hijacking Attacks for a 5G-Advanced Core Network," 2023 European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit), pp. 538–542, June 2023.
- **Seungchan Woo** and Jong-Hyouk Lee, "Proposal of Docker and Kubernetes Direction through the Event Timeline of Kubernetes," The 5th International Conference on Artificial Intelligence in Information and Communication (ICAIIC 2023), February 2023
- **Seungchan Woo**, Jaehyoung Park, Kyungmin Park, Jonghyun Kim and Jong-Hyouk Lee, "Privilege Escalation Attack Using Vulnerabilities in Pod Creation through Docker Local Registry in Kubernetes," KIISC Fall Conference, November 2022.
- **Seungchan Woo** and Jong-Hyouk Lee, "Analysis of Possible Security Threats in Kubernetes Environments," KIISC Fall Conference, November 2021.

Research Interests

- Cybersecurity standardization
- Anti-drone and counter-UAS system interoperability
- Drone and AI-based drone security
- 5G-Advanced and 6G network security
- Open RAN and UxNB security
- Blockchain, DLT, decentralized identifiers, and privacy protection
- Security-by-design and security management for emerging ICT systems

Awards

- Best Paper Award, Korea Institute of Information Security & Cryptology (KIISC), 2023.
- Chairman's Award, Side Channel Information Analysis Research Group, 2023.

Language

- Korean: Native
- English: Intermediate